

CSA 6503 - Generative Ai

CO2-AT2

20. PEFT Strategy Selection

1. Introduction

A company wants to customize a Large Language Model (LLM) using a **small domain-specific dataset** (for example, legal documents, medical reports, or company manuals). The organization must decide whether to use **Full Fine-Tuning** or **Parameter-Efficient Fine-Tuning (PEFT)** techniques such as **LoRA**.

2. Problem Analysis

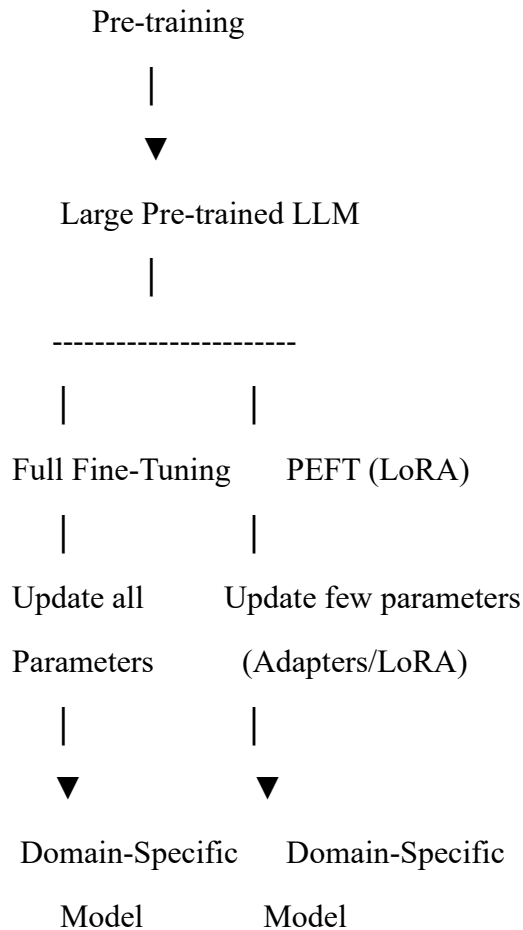
Scenario

- Small domain-specific dataset.
- Large pre-trained model.
- Limited GPU resources.
- Need good domain-specific performance.
- Fast deployment and low cost.

Challenges

- Full fine-tuning requires high-end GPUs.
- Large storage requirements.
- Risk of overfitting due to small dataset.
- High training cost.

3. LLM Architecture



4. Pre-training and Fine-Tuning

Pre-training

- Model is trained on billions of words.
- Learns grammar, reasoning, and general knowledge.

Examples:

- Llama
- Mistral
- Gemma

Fine-Tuning

The company adapts the model to its own domain.

Example:

General Model → Medical Chatbot

5. Transfer Learning

Transfer learning allows the company to reuse the knowledge learned during pre-training.

Instead of training from scratch,

General Knowledge

+

Small Medical Dataset

↓

Medical Assistant

Benefits:

- Less data required.
- Faster training.
- Better accuracy.

6. Full Fine-Tuning vs PEFT

Feature	Full Fine-Tuning	PEFT (LoRA)
Parameters Updated	All	Few
GPU Requirement	Very High	Low
Training Time	Long	Short
Storage	Large	Small
Cost	High	Low
Overfitting Risk	High	Low
Best For	Large datasets	Small datasets

7. Why PEFT is Better

Since the dataset is small,

PEFT offers

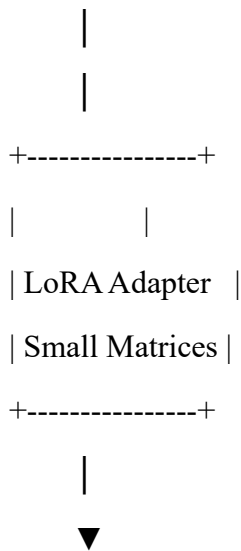
- Lower computational cost
- Faster training
- Less memory usage
- Easier deployment
- Similar performance to full fine-tuning

Therefore,

Recommended Method: PEFT using LoRA

8. LoRA Architecture

Frozen LLM Weights



Domain-Specific Output

Only the adapter weights are trained.

9. Prompt Engineering

Well-designed prompts improve model accuracy.

Example

Prompt

You are a medical assistant.

Answer only using the hospital guidelines.

Question:

What is diabetes?

10. Zero-shot, One-shot and Few-shot Prompting

Zero-shot

Translate:

Hello

Output:

Hola

One-shot

Example:

Dog → Animal

Cat →

Output:

Animal

Few-shot

Apple → Fruit

Carrot → Vegetable

Lion → Animal

Rose →

Output:

Flower

Few-shot prompting usually performs better for specialized tasks.

11. Reasoning-Oriented Prompting

Example

Solve step by step.

Question:

A company buys 10 laptops costing ₹40,000 each.

Find the total cost.

Output

$10 \times 40,000$

$= ₹4,00,000$

Useful for:

- Finance
- Mathematics
- Business

12. Structured Prompting

Example

Role:

Medical Expert

Task:

Explain diabetes.

Format:

Definition

Symptoms

Treatment

Prevention

Produces organized responses.

13. Function Calling

Example Workflow

User

|



LLM

|

Calls Function

|



Hospital Database

|

Returns Result

|



LLM Response

Example

User:

Show today's appointments.

Function

getAppointments()

14. Prompt Evaluation

Evaluate prompts using

- Accuracy
- Relevance
- Completeness
- Consistency
- Hallucination rate

Example

Poor Prompt

Tell me about AI.

Better Prompt

Explain Artificial Intelligence in 100 words with one example.

15. Security and Edge-Case Handling

Possible Risks

- Prompt injection
- Data leakage
- Hallucinations
- Unauthorized access

Solutions

- Input validation
- Output filtering
- Access control
- Secure APIs
- Human review for critical decisions

16. Advantages of PEFT

- Low GPU usage
- Low training cost
- Fast adaptation
- Easy deployment
- Supports multiple adapters
- High accuracy on small datasets

17. Limitations

- Slightly lower performance than full fine-tuning in some large-data scenarios.
- Adapter management becomes important when supporting many domains.

18. Recommended Workflow

Domain Dataset

|

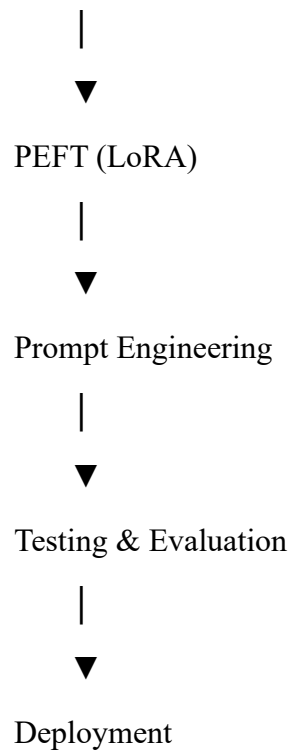


Data Cleaning

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Pre-trained LLM



19. Final Recommendation

For a company with a **small domain-specific dataset**, **PEFT using LoRA** is the best solution because it:

- Reduces computational cost.
- Requires fewer GPU resources.
- Trains faster.
- Avoids overfitting.
- Delivers strong domain-specific performance.

31. Hugging Face Deployment

1. Introduction

A research team wants greater control over model selection, customization, and deployment. They are considering using **Hugging Face** models instead of hosted LLM APIs.

2. Problem Analysis

Scenario

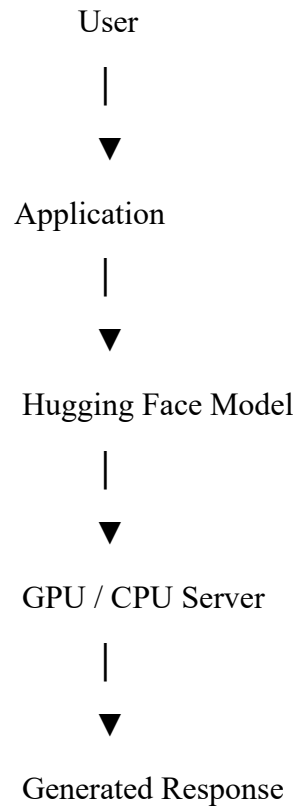
The team needs:

- Custom model selection
- Fine-tuning capability
- Local deployment
- Privacy
- Research flexibility

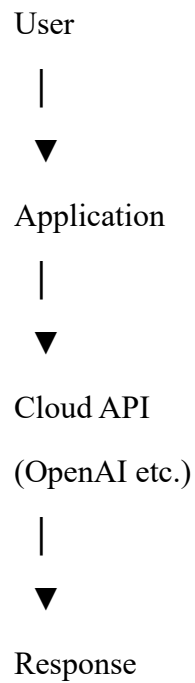
Challenges

- Hardware requirements
- Deployment complexity
- Maintenance
- Model optimization

3. LLM Architecture



Hosted API



4. Pre-training and Fine-Tuning

Hugging Face provides many pre-trained models.

Examples

- Llama
- Mistral
- Gemma
- Phi

Researchers can fine-tune them using domain-specific datasets.

5. Transfer Learning

Instead of building a model from scratch,

Pre-trained Model

|



Domain Dataset

|



Fine-Tuned Model

Benefits

- Faster development.
- Lower training cost.
- Better performance.

6. Hugging Face vs Hosted APIs

Feature	Hugging Face	Hosted API
Model Control	Full	Limited

Feature	Hugging Face	Hosted API
Fine-Tuning	Yes	Limited/Provider dependent
Offline Deployment	Yes	No
Custom Models	Yes	Limited
Privacy	High	Data sent to provider
Infrastructure	User manages	Provider manages
Cost	Hardware cost	Pay per request
Maintenance	High	Low

7. Advantages of Hugging Face

- Open-source ecosystem.
- Thousands of pre-trained models.
- Supports LoRA and PEFT.
- Works offline.
- Better privacy.
- Easy experimentation.

8. Challenges

- Requires GPUs for large models.
- Deployment complexity.
- Model updates.
- Security management.
- Infrastructure cost.

9. Prompt Engineering

Example

Role:

Financial Advisor

Task:

Explain investment risk.

Limit:

150 words

Produces consistent outputs.

10. Zero-shot / One-shot / Few-shot

Zero-shot

Summarize this article.

One-shot

Example:

Positive → Happy

Negative →

Few-shot

Positive → Happy

Negative → Sad

Angry → Emotion

Excited →

11. Reasoning-Oriented Prompting

Explain your reasoning step by step before giving the answer.

Useful for

- Coding
- Mathematics
- Scientific analysis

12. Structured Prompting

Role:

Research Assistant

Task:

Summarize the paper

Output

1. Objective

2. Method

3. Results

4. Conclusion

13. Function Calling

Workflow

User

|



LLM

|

Calls API

(Database/Search)

|



Returns Data

|



LLM Response

Example

searchResearchPaper()

14. PEFT (LoRA)

Researchers can fine-tune Hugging Face models using LoRA.

Benefits

- Faster training
- Lower memory
- Smaller checkpoints
- Better customization

15. LLM API Selection

Requirement	Recommended Choice
Fast development	Hosted API
Maximum customization	Hugging Face
Research	Hugging Face
Enterprise privacy	Hugging Face (self-hosted)
Small startup	Hosted API

16. Prompt Evaluation

Evaluation Metrics

- Accuracy
- Relevance
- Latency
- Consistency
- Hallucination rate
- User satisfaction

17. Security and Edge-Case Handling

Potential Issues

- Prompt injection
- Unsafe outputs
- Hallucinations
- Unauthorized access

Solutions

- Prompt validation
- Content moderation

- Authentication
- Rate limiting
- Logging and monitoring
- Human review for sensitive use cases

18. Deployment Workflow

Research Dataset

|



Choose Hugging Face Model

|



PEFT (LoRA)

|



Prompt Engineering

|



Testing

|



Deploy on GPU Server

|



User Application

19. Applications

- Healthcare assistants
- Legal document analysis
- Financial advisory systems
- Educational chatbots
- Research assistants

20. Final Recommendation

For a research team that requires **greater control, customization, and privacy**, a **self-hosted Hugging Face deployment** is the most suitable choice. It supports:

- Full model selection and customization.
- PEFT (LoRA) for efficient fine-tuning.
- Strong transfer learning capabilities.
- Flexible prompt engineering.
- Function calling with external tools and databases.
- Offline or private deployments for sensitive data.